Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John Brasil Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
2613 S Mitchell RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X044-X020-X031-XXXX X058-X016-X004-XXXX X058-X150-X006-XXXX X058-X160-X004-XXXX X058-X160-X011-XXXX X058-X290-X014-XXXX X058-X290-X023-XXXX

B. OPERATORS

Brasil, John

Operator name: Brasil, John

Telephone no.: (209) 632-7867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2613 S Mitchell RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Brasil, John

Legal owner name: Brasil, John

Telephone no.: (209) 632-7867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2613 S Mitchell RD	Turlock	CA	95380

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	132	498	478	273	0
Number under roof	1,155	0	0	0	0	0
Maximum number	1,155	132	498	478	273	0
Average number	1,155	132	498	478	273	0
Avg live weight (lbs)	1,400	1,500	800	600		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 42,458.49 *tons per reporting period*

Total nitrogen from manure: 526,888.22 *lbs per reporting period*

After ammonia losses (30% loss applied): 368,821.75 *lbs per reporting period*

Total phosphorus from manure: 87,048.71 *lbs per reporting period*

Total potassium from manure: 224,987.33 *lbs per reporting period*

Total salt from manure: 574,185.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,580,275 *gallons*

Total nitrogen generated: 73,706.99 *lbs*

Total phosphorus generated: 11,096.43 *lbs*

Total potassium generated: 143,634.39 *lbs*

Total salt generated: 846,332.58 *lbs*

	16,422,275	*gallons applied*
+	0	*gallons exported*
-	3,842,000	*gallons imported*
=	12,580,275	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW-Barn	Ground water
Domestic Well - DW-House	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
04/17/2023	Process wastewater *LWI-JDB*	1,672,000.00 *gal*	519.90	88.50	885.00		3,650
08/20/2023	Process wastewater *LWI-JDB*	2,170,000.00 *gal*	519.90	88.50	885.00		3,650

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	16,668.77	2,837.44	28,374.42	117,024.44
Total imports for all materials	16,668.77	2,837.44	28,374.42	117,024.44

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	Corral solids	1,600.00 *ton*	Dry-weight	11.1		13,700.00	5,400.00	18,200.00		0.00
10/16/2023	Corral solids	1,200.00 *ton*	Dry-weight	6.4		26,300.00	7,500.00	22,800.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	98,054.08	32,209.92	102,993.28	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	98,054.08	32,209.92	102,993.28	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Clayton	40	40	2	both	X044-X020-X031-XXXX
Dairy East	40	40	2	process wastewater	X058-X150-X006-XXXX
Dairy West	15	15	2	process wastewater	X058-X016-X004-XXXX
Goat Ranch	40	40	2	process wastewater	X058-X290-X023-XXXX
Linwood	40	40	2	process wastewater	X058-X160-X011-XXXX
Moffett	18	18	2	process wastewater	X058-X290-X014-XXXX
Santos	79	79	0	none	X044-X043-X007-XXXX
Totals for areas that were used for application	193	193	12		
Totals for areas that were not used for application	79	79	0		
Land application area totals	272	272	12		

B. CROPS AND HARVESTS

Clayton

Field name: Clayton

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	679.20 *ton*	Dry-weight		82.6	19,000.00	4,900.00	34,000.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.98	112.27	28.95	200.91	726.81

Clayton

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,075.44 *ton*	Dry-weight		68.0	13,300.00	2,900.00	12,800.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.89	228.85	49.90	220.25	808.73

Dairy East

Field name: Dairy East

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	678.92 *ton*	Dry-weight		83.2	19,800.00	5,500.00	44,500.00		15.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.97	112.92	31.37	253.78	889.66

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,224.31 *ton*	Dry-weight		68.6	13,000.00	2,500.00	12,700.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.61	249.88	48.05	244.12	903.42

Dairy West

Field name: Dairy West

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	269.52 *ton*	Dry-weight		83.1	20,300.00	5,100.00	33,600.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.97	123.29	30.97	204.06	734.86

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	481.32 *ton*	Dry-weight		68.2	12,100.00	2,300.00	13,600.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.09	246.94	46.94	277.55	1,020.40

Goat Ranch

Field name: Goat Ranch

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	719.10 *ton*	Dry-weight		84.1	19,100.00	4,900.00	41,600.00		15.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.98	109.19	28.01	237.82	891.83

Goat Ranch

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,051.23 *ton*	Dry-weight		65.1	13,700.00	2,500.00	10,300.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.28	251.31	45.86	188.94	862.17

Linwood

Field name: Linwood

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	671.23 *ton*	Dry-weight		82.5	19,400.00	5,500.00	42,200.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.78	113.94	32.30	247.85	816.38

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,197.55 *ton*	Dry-weight		65.5	12,500.00	2,400.00	12,200.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.94	258.22	49.58	252.02	929.60

Moffett

Field name: Moffett

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	304.88 *ton*	Dry-weight		83.6	21,700.00	5,600.00	44,400.00		16.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.94	120.56	31.11	246.67	905.56

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	533.24 *ton*	Dry-weight		66.7	12,500.00	3,000.00	13,000.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.62	246.62	59.19	256.49	966.76

NUTRIENT BUDGET

A. LAND APPLICATIONS

Clayton - 11/30/2022: Oats, silage-soft dough

Field name: Clayton

Crop: Oats, silage-soft dough — Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI-JBD	Process wastewater	67.25	15.46	178.26	723.64	846,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.00	4,074,675.00 *gal*
Application event totals		67.25	15.46	178.26	757.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI-JBD	Process wastewater	65.66	15.10	174.05	706.53	826,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.28	4,108,350.00 *gal*
Application event totals		65.66	15.10	174.05	740.81	

Clayton - 05/22/2023: Corn, silage

Field name: Clayton

Crop: Corn, silage — Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	79.17	31.20	105.17	0.00	130.00 *ton*
Application event totals		79.17	31.20	105.17	0.00	

Clayton - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	22.26	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	22.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI - JBD	Process wastewater	56.62	9.64	96.38	397.49	522,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.12	5,286,975.00 *gal*
Application event totals		56.62	9.64	96.38	441.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.05	5,758,425.00 *gal*
Application event totals		0.00	0.00	0.00	48.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI - JBD	Process wastewater	55.32	9.42	94.16	388.36	510,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		55.32	9.42	94.16	434.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI - JBD	Process wastewater	62.91	10.71	107.09	441.66	580,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	5,421,675.00 *gal*
Application event totals		62.91	10.71	107.09	486.90	

Clayton - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LWI - JBD	Process wastewater	60.52	10.30	103.03	424.91	558,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.68	5,354,325.00 *gal*
Application event totals		60.52	10.30	103.03	469.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.68	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	44.68	

Dairy East - 11/30/2022: Oats, silage-soft dough

Field name: Dairy East

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.11	10.98	144.58	1,021.22	550,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.68	5,354,325.00 *gal*
Application event totals		77.11	10.98	144.58	1,065.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.30	10.58	139.32	984.08	530,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	5,421,675.00 *gal*
Application event totals		74.30	10.58	139.32	1,029.33	

Dairy East - 05/22/2023: Corn, silage

Dairy East - 05/22/2023: Corn, silage

Field name: Dairy East

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	45.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.93	14.23	162.94	803.21	550,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.87	5,017,575.00 *gal*
Application event totals		86.93	14.23	162.94	845.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.68	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	44.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.77	13.71	157.01	774.00	530,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		83.77	13.71	157.01	819.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.61	13.19	151.09	744.79	510,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	5,421,675.00 *gal*
Application event totals		80.61	13.19	151.09	790.04	

Dairy East - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.97	14.24	163.02	803.61	550,275.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	4,950,225.00 *gal*
Application event totals		86.97	14.24	163.02	844.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.15	5,051,250.00 *gal*
Application event totals		0.00	0.00	0.00	42.15	

Dairy West - 11/30/2022: Oats, silage-soft dough

Field name: Dairy West

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.99	12.25	161.23	1,138.81	230,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	58.45	2,626,650.00 *gal*
Application event totals		85.99	12.25	161.23	1,197.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.51	11.18	147.21	1,039.79	210,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	66.69	2,997,075.00 *gal*
Application event totals		78.51	11.18	147.21	1,106.48	

Dairy West - 05/22/2023: Corn, silage

Dairy West - 05/22/2023: Corn, silage

Field name: Dairy West

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.20	2,660,325.00 *gal*
Application event totals		0.00	0.00	0.00	59.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	57.70	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	57.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.90	16.50	217.30	1,534.92	310,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.20	2,525,625.00 *gal*
Application event totals		115.90	16.50	217.30	1,591.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.16	15.97	210.29	1,485.41	300,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	58.45	2,626,650.00 *gal*
Application event totals		112.16	15.97	210.29	1,543.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.20	2,660,325.00 *gal*
Application event totals		0.00	0.00	0.00	59.20	

Dairy West - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.42	15.44	203.28	1,435.90	290,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	57.70	2,592,975.00 *gal*
Application event totals		108.42	15.44	203.28	1,493.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.20	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	56.20	

Goat Ranch - 11/30/2022: Oats, silage-soft dough

Field name: Goat Ranch

Crop: Oats, silage-soft dough Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.71	10.78	141.95	1,002.65	540,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.75	4,882,875.00 *gal*
Application event totals		75.71	10.78	141.95	1,043.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.10	9.98	131.43	928.38	500,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	4,546,125.00 *gal*
Application event totals		70.10	9.98	131.43	966.32	

Goat Ranch - 05/22/2023: Corn, silage

Goat Ranch - 05/22/2023: Corn, silage

Field name: Goat Ranch

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.55	6,297,225.00 *gal*
Application event totals		0.00	0.00	0.00	52.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.62	5,825,775.00 *gal*
Application event totals		0.00	0.00	0.00	48.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.20	16.44	239.96	956.88	710,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.43	6,162,525.00 *gal*
Application event totals		122.20	16.44	239.96	1,008.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.32	15.52	226.44	902.97	670,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.86	6,095,175.00 *gal*
Application event totals		115.32	15.52	226.44	953.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.18	5,893,125.00 *gal*
Application event totals		0.00	0.00	0.00	49.18	

Goat Ranch - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.43	14.59	212.92	849.06	630,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.99	6,229,875.00 *gal*
Application event totals		108.43	14.59	212.92	901.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.86	6,095,175.00 *gal*
Application event totals		0.00	0.00	0.00	50.86	

Linwood - 11/30/2022: Oats, silage-soft dough

Field name: Linwood

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.41	9.71	114.38	1,156.62	560,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	4,950,225.00 *gal*
Application event totals		76.41	9.71	114.38	1,197.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.71	10.78	141.95	1,002.65	540,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.75	4,882,875.00 *gal*
Application event totals		75.71	10.78	141.95	1,043.40	

Linwood - 05/22/2023: Corn, silage

Linwood - 05/22/2023: Corn, silage

Field name: Linwood

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	45.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	45.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	124.86	20.44	234.04	1,153.70	790,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.84	5,253,300.00 *gal*
Application event totals		124.86	20.44	234.04	1,197.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.38	18.88	216.26	1,066.07	730,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.75	4,882,875.00 *gal*
Application event totals		115.38	18.88	216.26	1,106.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	41.31	

Linwood - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.15	14.97	197.15	1,392.57	750,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		105.15	14.97	197.15	1,438.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.12	5,286,975.00 *gal*
Application event totals		0.00	0.00	0.00	44.12	

Moffett - 11/30/2022: Oats, silage-soft dough

Field name: Moffett

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.12	11.98	157.72	1,114.06	270,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	60.57	3,266,475.00 *gal*
Application event totals		84.12	11.98	157.72	1,174.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.89	11.09	146.04	1,031.53	250,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.33	2,929,725.00 *gal*
Application event totals		77.89	11.09	146.04	1,085.86	

Moffett - 05/22/2023: Corn, silage

Moffett - 05/22/2023: Corn, silage

Field name: Moffett

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.83	3,064,425.00 *gal*
Application event totals		0.00	0.00	0.00	56.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.08	2,862,375.00 *gal*
Application event totals		0.00	0.00	0.00	53.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	121.50	17.30	227.82	1,609.19	390,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.33	2,929,725.00 *gal*
Application event totals		121.50	17.30	227.82	1,663.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.04	15.53	204.45	1,444.15	350,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.83	3,064,425.00 *gal*
Application event totals		109.04	15.53	204.45	1,500.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.08	2,862,375.00 *gal*
Application event totals		0.00	0.00	0.00	53.08	

Moffett - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.93	15.08	198.61	1,402.89	340,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.33	2,929,725.00 *gal*
Application event totals		105.93	15.08	198.61	1,457.22	

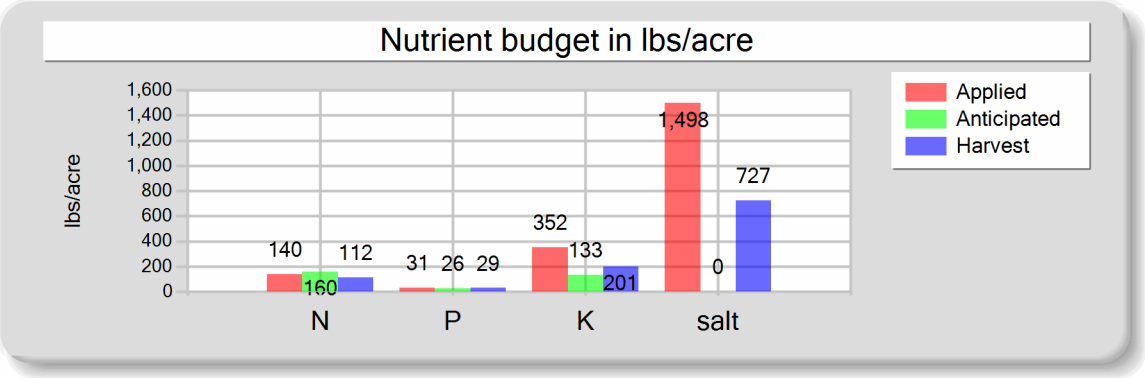
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.71	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	48.71	

B. NUTRIENT BUDGET

Clayton - 11/30/2022: Oats, silage-soft dough

Field name: Clayton | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	132.90	30.56	352.31	1,430.17
Fresh water	0.00	0.00	0.00	68.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.90	30.56	352.31	1,498.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.27	28.95	200.91	726.81
Nutrient balance	27.63	1.60	151.40	771.64
Applied to removed ratio	1.25	1.06	1.75	2.06

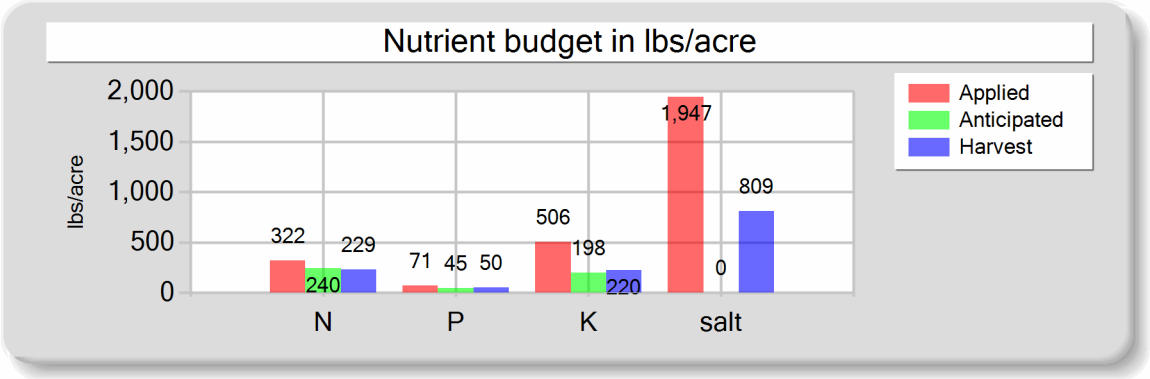
Fresh water applied	
8,183,025.00	*gallons*
301.35	*acre-inches*
7.53	*inches/acre*

Process wastewater applied	
1,672,000.00	*gallons*
61.57	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Clayton - 05/22/2023: Corn, silage

Field name: Clayton | Crop: Corn, silage | Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.17	31.20	105.17	0.00
Process wastewater	235.37	40.07	400.65	1,652.41
Fresh water	0.00	0.00	0.00	294.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	321.53	71.27	505.82	1,947.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	228.85	49.90	220.25	808.73
Nutrient balance	92.68	21.37	285.57	1,138.53
Applied to removed ratio	1.40	1.43	2.30	2.41

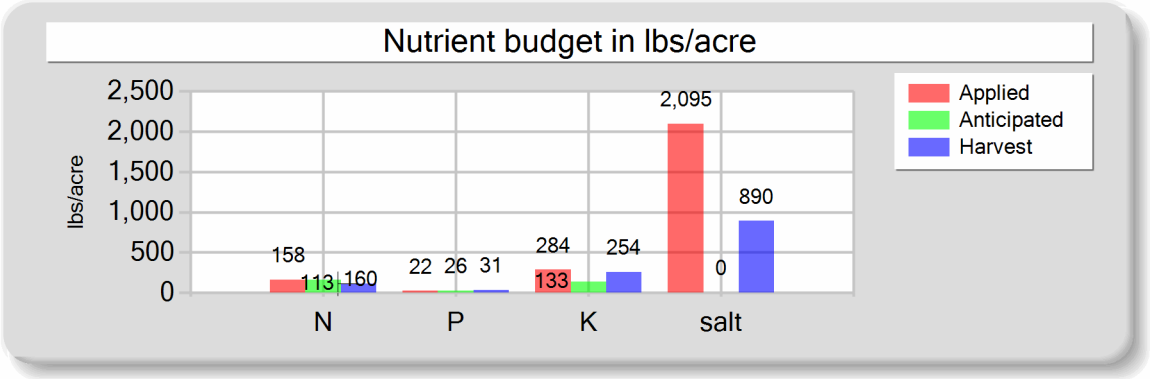
Fresh water applied	
38,086,425.00	*gallons*
1,402.59	*acre-inches*
35.06	*inches/acre*

Process wastewater applied	
2,170,000.00	*gallons*
79.91	*acre-inches*
2.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy East - 11/30/2022: Oats, silage-soft dough

Field name: Dairy East | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	151.41	21.56	283.90	2,005.30
Fresh water	0.00	0.00	0.00	89.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.41	21.56	283.90	2,095.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.92	31.37	253.78	889.66
Nutrient balance	45.49	-9.80	30.12	1,205.57
Applied to removed ratio	1.40	0.69	1.12	2.36

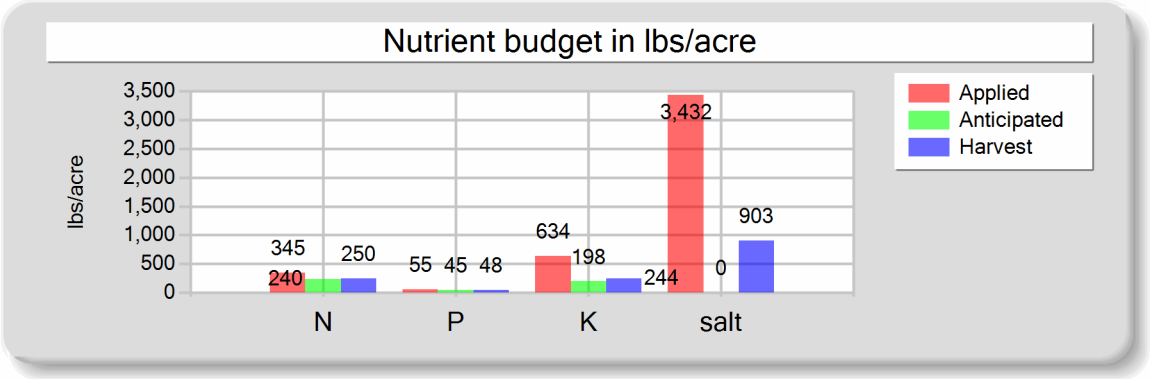
Fresh water applied	
10,776,000.00	*gallons*
396.84	*acre-inches*
9.92	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy East - 05/22/2023: Corn, silage

Field name: Dairy East | Crop: Corn, silage | Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	338.28	55.37	634.05	3,125.60
Fresh water	0.00	0.00	0.00	306.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.28	55.37	634.05	3,432.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.88	48.05	244.12	903.42
Nutrient balance	95.40	7.31	389.94	2,529.06
Applied to removed ratio	1.38	1.15	2.60	3.80

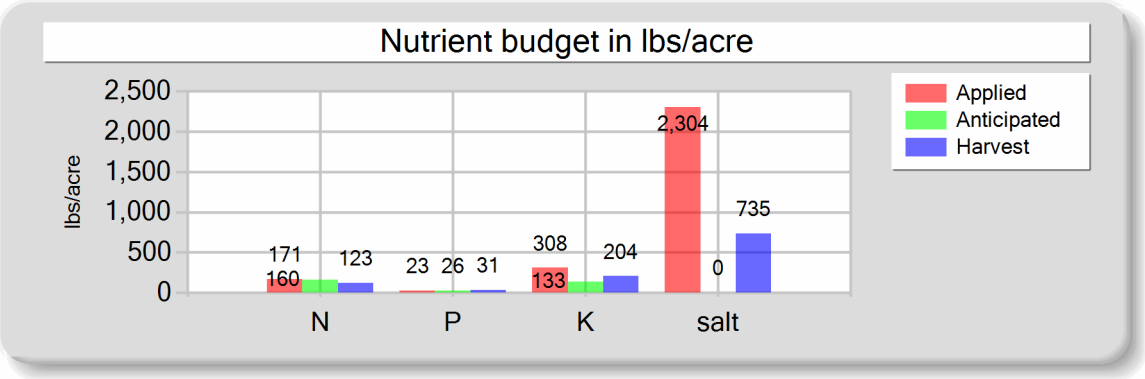
Fresh water applied	
36,773,100.00	*gallons*
1,354.23	*acre-inches*
33.86	*inches/acre*

Process wastewater applied	
2,140,275.00	*gallons*
78.82	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy West - 11/30/2022: Oats, silage-soft dough

Field name: Dairy West | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.50	23.43	308.43	2,178.60
Fresh water	0.00	0.00	0.00	125.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	171.50	23.43	308.43	2,303.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	123.29	30.97	204.06	734.86
Nutrient balance	48.21	-7.55	104.37	1,568.89
Applied to removed ratio	1.39	0.76	1.51	3.13

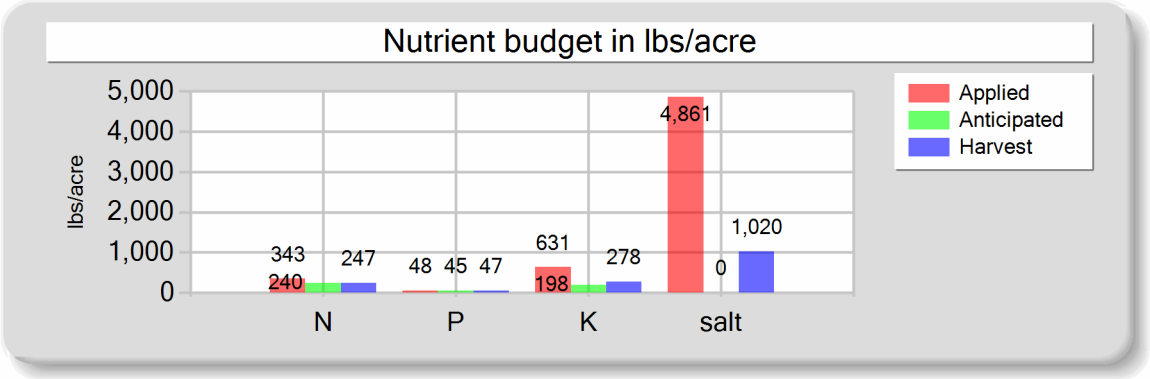
Fresh water applied	
5,623,725.00	*gallons*
207.10	*acre-inches*
13.81	*inches/acre*

Process wastewater applied	
440,000.00	*gallons*
16.20	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy West - 05/22/2023: Corn, silage

Field name: Dairy West | Crop: Corn, silage | Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	336.47	47.92	630.88	4,456.23
Fresh water	0.00	0.00	0.00	404.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.47	47.92	630.88	4,860.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	246.94	46.94	277.55	1,020.40
Nutrient balance	96.53	0.98	353.33	3,840.50
Applied to removed ratio	1.39	1.02	2.27	4.76

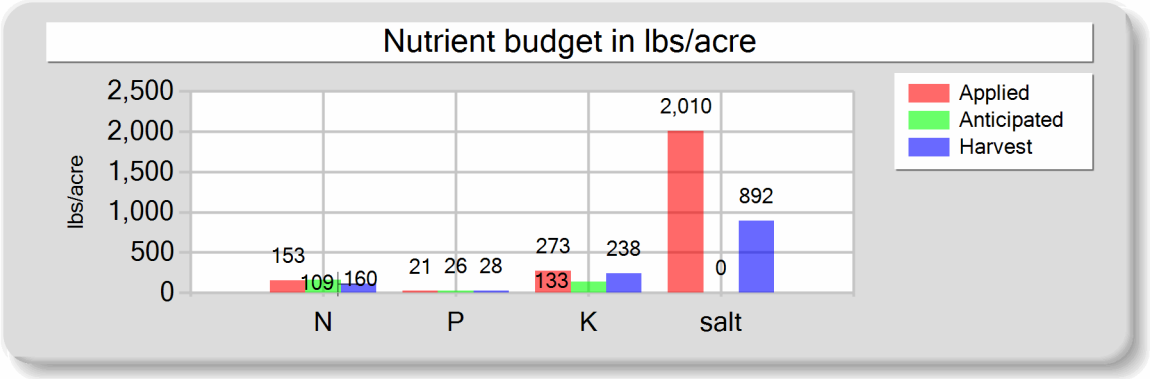
Fresh water applied	
18,184,500.00	*gallons*
669.67	*acre-inches*
44.64	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Goat Ranch - 11/30/2022: Oats, silage-soft dough

Field name: Goat Ranch | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	145.80	20.76	273.38	1,931.03
Fresh water	0.00	0.00	0.00	78.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.80	20.76	273.38	2,009.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	109.19	28.01	237.82	891.83
Nutrient balance	43.61	-7.25	35.56	1,117.89
Applied to removed ratio	1.40	0.74	1.15	2.25

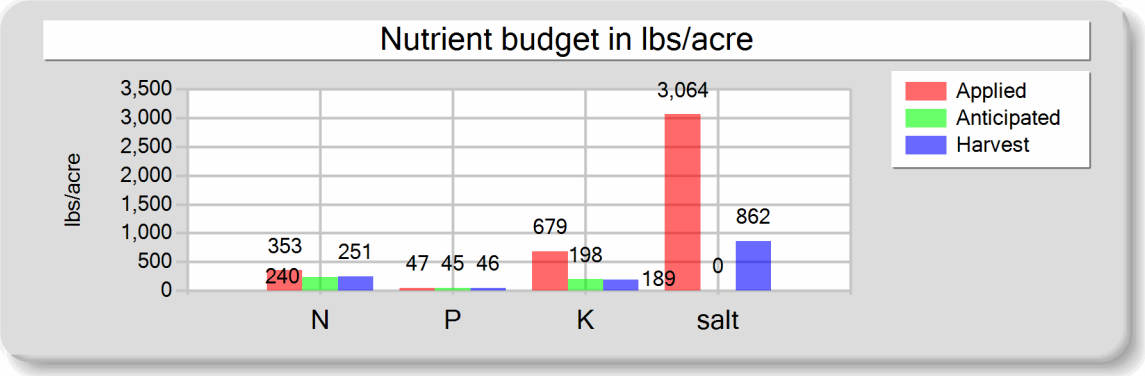
Fresh water applied	
9,429,000.00	*gallons*
347.24	*acre-inches*
8.68	*inches/acre*

Process wastewater applied	
1,040,000.00	*gallons*
38.30	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Goat Ranch - 05/22/2023: Corn, silage

Field name: Goat Ranch Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	345.95	46.55	679.32	2,708.91
Fresh water	0.00	0.00	0.00	355.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	352.95	46.55	679.32	3,064.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	251.31	45.86	188.94	862.17
Nutrient balance	101.64	0.69	490.38	2,202.23
Applied to removed ratio	1.40	1.01	3.60	3.55

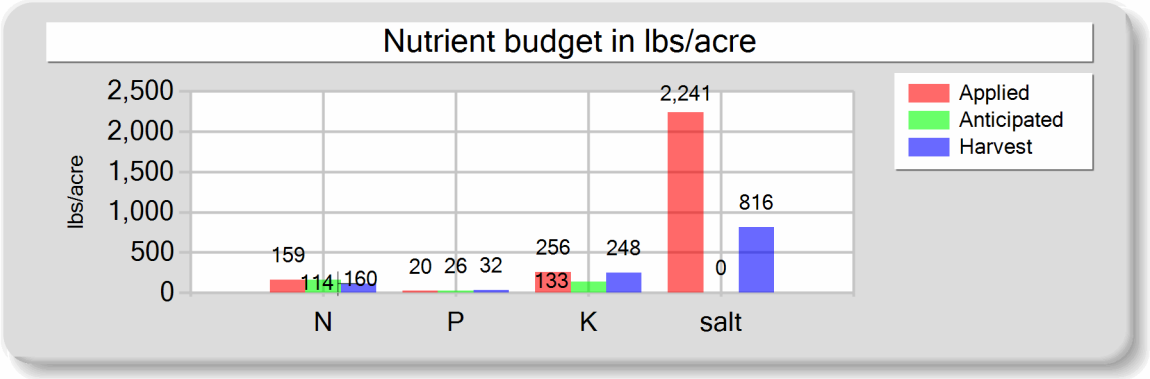
Fresh water applied	
42,598,875.00	*gallons*
1,568.77	*acre-inches*
39.22	*inches/acre*

Process wastewater applied	
2,010,000.00	*gallons*
74.02	*acre-inches*
1.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Linwood - 11/30/2022: Oats, silage-soft dough

Field name: Linwood | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	152.11	20.49	256.33	2,159.27
Fresh water	0.00	0.00	0.00	82.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.11	20.49	256.33	2,241.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	113.94	32.30	247.85	816.38
Nutrient balance	45.17	-11.81	8.47	1,424.94
Applied to removed ratio	1.40	0.63	1.03	2.75

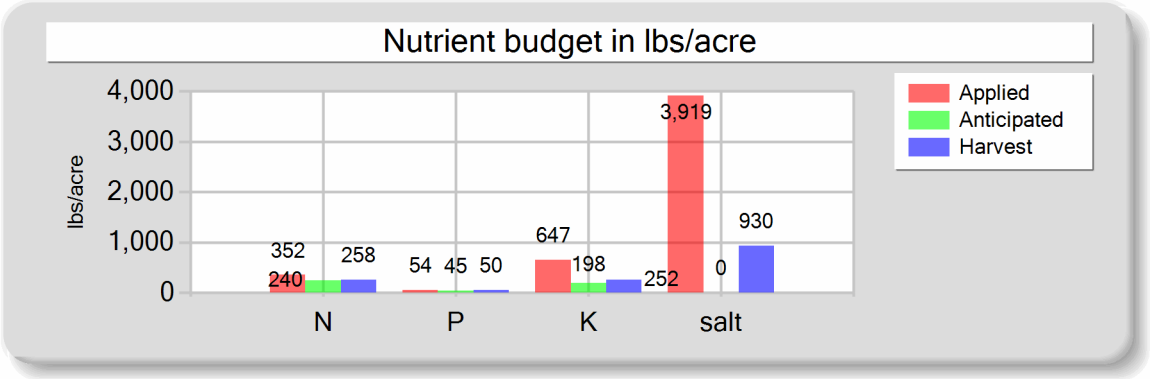
Fresh water applied	
9,833,100.00	*gallons*
362.12	*acre-inches*
9.05	*inches/acre*

Process wastewater applied	
1,100,000.00	*gallons*
40.51	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Linwood - 05/22/2023: Corn, silage

Field name: Linwood Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	345.39	54.30	647.45	3,612.34
Fresh water	0.00	0.00	0.00	306.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	352.39	54.30	647.45	3,919.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.22	49.58	252.02	929.60
Nutrient balance	94.17	4.72	395.42	2,989.62
Applied to removed ratio	1.36	1.10	2.57	4.22

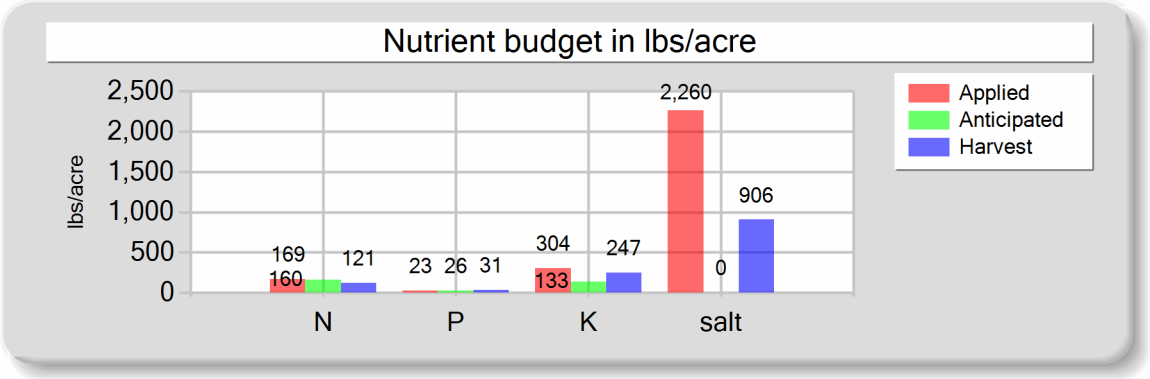
Fresh water applied	
36,773,100.00	*gallons*
1,354.23	*acre-inches*
33.86	*inches/acre*

Process wastewater applied	
2,270,000.00	*gallons*
83.60	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Moffett - 11/30/2022: Oats, silage-soft dough

Field name: Moffett | Crop: Oats, silage-soft dough | Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.00	23.07	303.76	2,145.59
Fresh water	0.00	0.00	0.00	114.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.00	23.07	303.76	2,260.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	120.56	31.11	246.67	905.56
Nutrient balance	48.45	-8.04	57.09	1,354.94
Applied to removed ratio	1.40	0.74	1.23	2.50

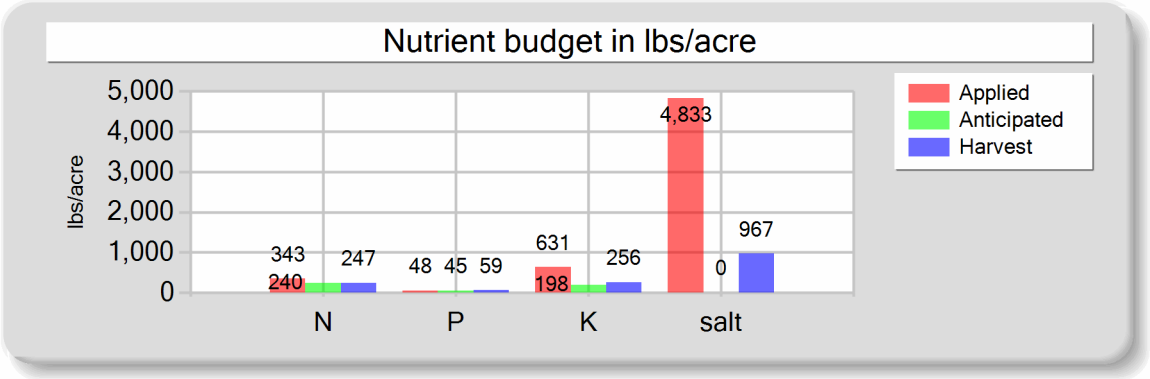
Fresh water applied	
6,196,200.00	*gallons*
228.18	*acre-inches*
12.68	*inches/acre*

Process wastewater applied	
520,000.00	*gallons*
19.15	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Moffett - 05/22/2023: Corn, silage

Field name: Moffett | Crop: Corn, silage | Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	336.47	47.92	630.88	4,456.23
Fresh water	0.00	0.00	0.00	377.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.47	47.92	630.88	4,833.42
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	246.62	59.19	256.49	966.76
Nutrient balance	96.85	-11.27	374.39	3,866.65
Applied to removed ratio	1.39	0.81	2.46	5.00

Fresh water applied	
20,339,700.00	*gallons*
749.04	*acre-inches*
41.61	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 06/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	5,400.00	18,200.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,300.00	7,500.00	22,800.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,770.00	700.00	0.00	1.50	666.00	1,470.00								12,800.00	4,300
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

LWI - JDB

Sample and source description: LWI - JDB

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	271.00	110.00	0.00	110.00	87.60	1,010.00								5,410.00	4,100
DL	1.00	0.50	1.00	0.50	0.10	1.00								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	824.00	394.00	0.00	1.00	111.00	1,620.00								12,200.00	6,460
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

LWI - JDB

Sample and source description: LWI - JDB

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	519.00	387.00	0.00	0.90	88.50	885.00								8,060.00	3,650
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	757.00	443.00	0.00	0.60	124.00	1,420.00								11,600.00	7,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	672.00	229.00	0.00	0.00	95.70	1,260.00								7,990.00	8,900
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW-Barn

DW - Barn

Sample description: DW - Barn

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			23.40								1,260.00	
DL			0.10								10.00	

Domestic Well - DW-House

DW - House

Sample description: DW - House

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			24.60								1,830.00	
DL			0.10								10.00	

Irrigation Water - Canal

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								32.80	40
DL	1.00		0.10								10.00	10

D. SOIL ANALYSES

Clayton

Clayton

Sample and source description: Clayton

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			87.00				
DL			2.00				

Dairy East

Dairy East

Sample and source description: Dairy East

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			155.00				
DL			2.00				

Dairy West

Dairy West

Dairy West

Sample and source description: Dairy West

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			315.00				
DL			2.00				

Linwood

Linwood

Sample and source description: Linwood

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			189.00				
DL			2.00				

Moffett

Moffett

Sample and source description: Moffett

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Clayton - 11/30/2022: Oats, silage-soft dough

Clayton - 11/30/2022: Oats, silage-soft dough

Clayton

Sample and source description: Clayton

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	4,900.00	34,000.00		12.30
DL	10.00	10.00	10.00		10.00

Clayton - 05/22/2023: Corn, silage

Clayton

Sample and source description: Clayton

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,900.00	12,800.00		4.70
DL	100.00	100.00	100.00		0.01

Dairy East - 11/30/2022: Oats, silage-soft dough

Dairy East

Sample and source description: Dairy East

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	5,500.00	44,500.00		15.60
DL	10.00	10.00	10.00		10.00

Dairy East - 05/22/2023: Corn, silage

Dairy East

Sample and source description: Dairy East

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,500.00	12,700.00		4.70
DL	100.00	100.00	100.00		0.01

Dairy West - 11/30/2022: Oats, silage-soft dough

Dairy West

Sample and source description: Dairy West

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	5,100.00	33,600.00		12.10
DL	10.00	10.00	10.00		10.00

Dairy West - 05/22/2023: Corn, silage

Dairy West

Sample and source description: Dairy West

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,300.00	13,600.00		5.00
DL	100.00	100.00	100.00		0.01

Goat Ranch - 11/30/2022: Oats, silage-soft dough

Goat Ranch

Sample and source description: Goat Ranch

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	4,900.00	41,600.00		15.60
DL	10.00	10.00	10.00		10.00

Goat Ranch - 05/22/2023: Corn, silage

Goat Ranch

Sample and source description: Goat Ranch

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,500.00	10,300.00		4.70
DL	100.00	100.00	100.00		0.01

Linwood - 11/30/2022: Oats, silage-soft dough

Linwood

Sample and source description: Linwood

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	5,500.00	42,200.00		13.90
DL	10.00	10.00	10.00		10.00

Linwood - 05/22/2023: Corn, silage

Linwood

Sample and source description: Linwood

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,400.00	12,200.00		4.50
DL	100.00	100.00	100.00		0.01

Moffett - 11/30/2022: Oats, silage-soft dough

Moffett

Sample and source description: Moffett

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	5,600.00	44,400.00		16.30
DL	10.00	10.00	10.00		10.00

Moffett - 05/22/2023: Corn, silage

Moffett

Sample and source description: Moffett

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	3,000.00	13,000.00		4.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

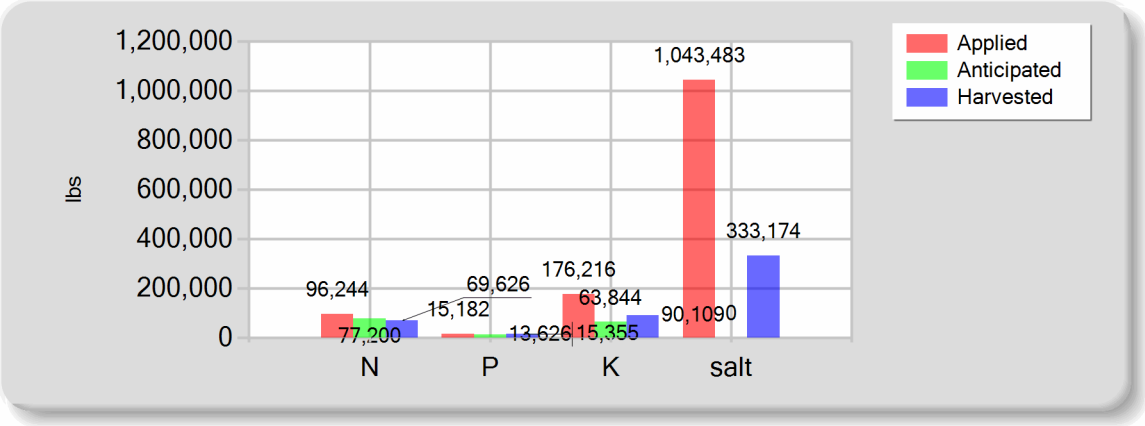
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

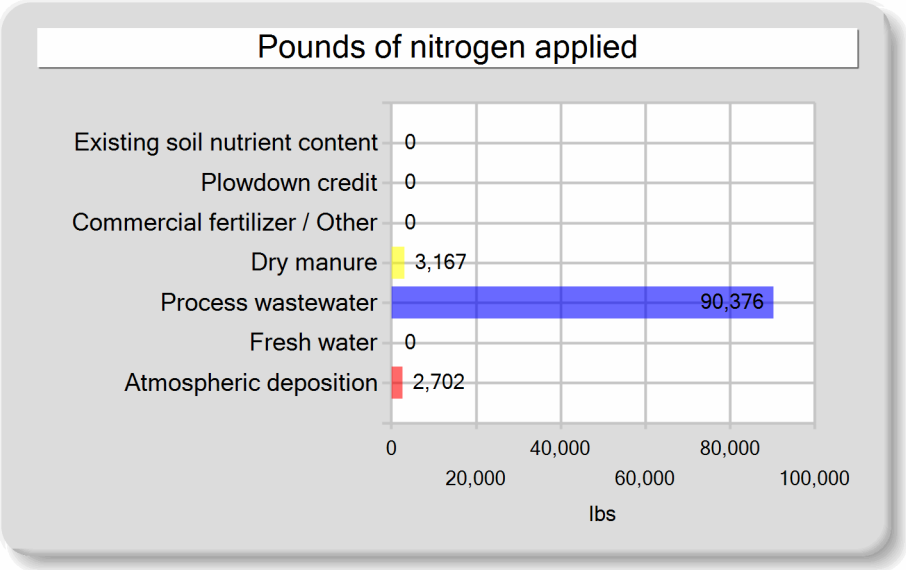
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

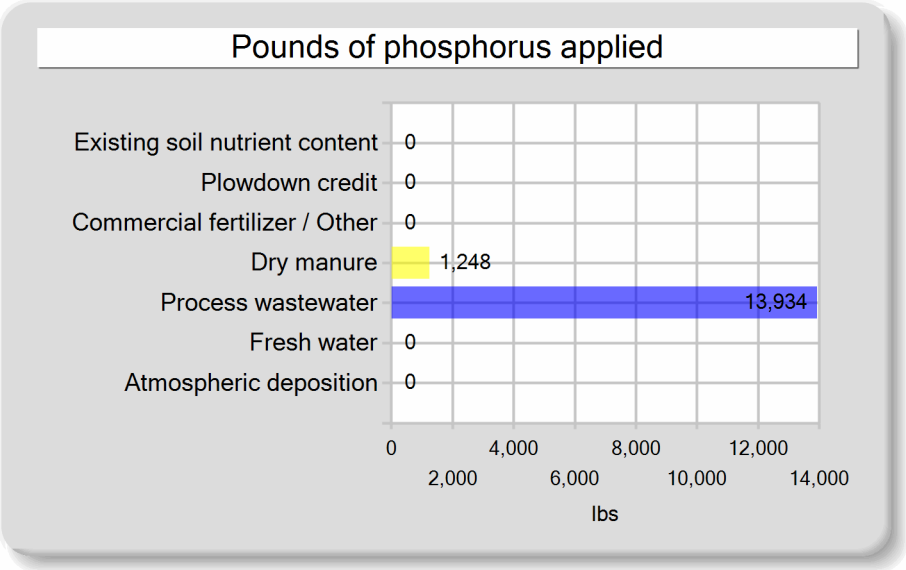
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	3,166.62	1,248.16	4,206.75	0.00
Process wastewater	90,375.76	13,933.87	172,008.81	963,357.02
Fresh water	0.00	0.00	0.00	80,126.20
Atmospheric deposition	2,702.00	0.00	0.00	0.00
Total nutrients applied	96,244.38	15,182.03	176,215.56	1,043,483.22
Anticipated crop nutrient removal	77,200.00	13,625.80	63,844.40	0.00
Actual crop nutrient removal	69,626.25	15,355.25	90,108.63	333,174.41
Nutrient balance	26,618.13	-173.22	86,106.93	710,308.81
Applied to removed ratio	1.38	0.99	1.96	3.13

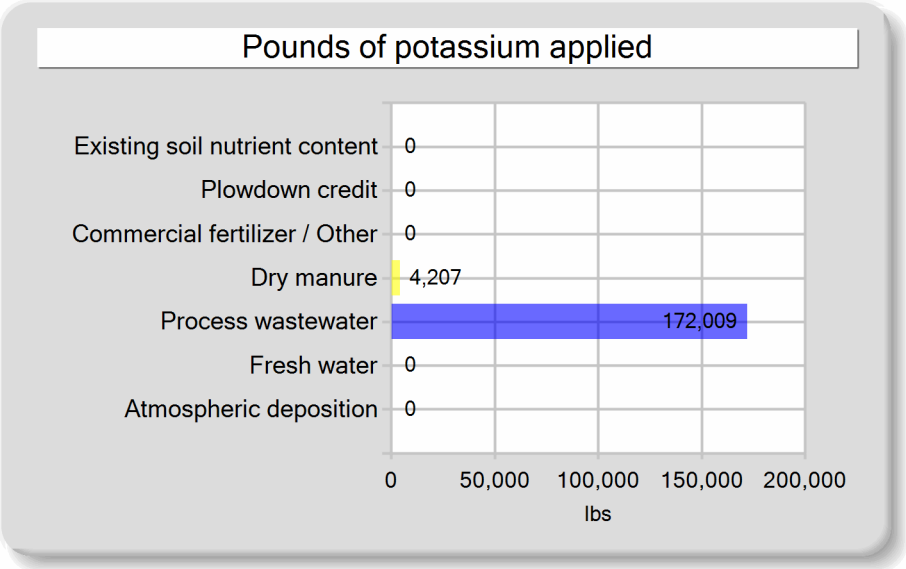
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

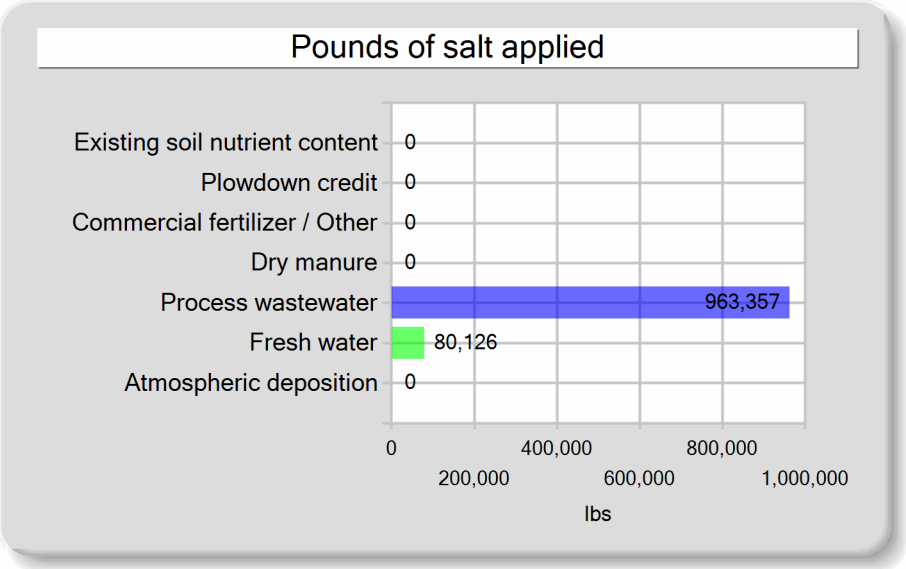


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Brasil	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/28/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **John Brasil**

Name of Dairy Facility: **John Brasil Dairy #1**

Facility Address: **2613 S. Mitchell Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **John Brasil (209) 652-3681**

Manure Hauler Information:

Name of Hauling Company/Person: **John Brasil Dairy #1**

Address of Hauling Company /Person: **2613 S. Mitchell Road, Turlock, CA 95380**

Contact Person and Phone Number: **John Brasil (209) 652-3681**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
JB Farms	8400 W. Linwood Avenue	Turlock	95380	(209) 656-9005

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
9730 Bradbury Road	Turlock	95380	

Dates Hauled: **5/22/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,600** Tons

Manure Solids Content (if amount reported in tons): **88.9%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 5/22/2023

Hauler's Signature: _____ Date: 5/22/2023

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **John Brasil**

Name of Dairy Facility: **John Brasil Dairy #1**

Facility Address: **2613 S. Mitchell Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **John Brasil (209) 652-3681**

Manure Hauler Information:

Name of Hauling Company/Person: **John Brasil Dairy #1**

Address of Hauling Company /Person: **2613 S. Mitchell Road, Turlock, CA 95380**

Contact Person and Phone Number: **John Brasil (209) 652-3681**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
JB Farms	8400 W. Linwood Avenue	Turlock	95380	(209) 656-9005

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
9730 Bradbury Road	Turlock	95380	

Dates Hauled: **10/16/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,200** Tons

Manure Solids Content (if amount reported in tons): **93.6%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 10/16/2023

Hauler's Signature: _____ Date: 10/16/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #1

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0684-01	DW - Barn	Ag Water	T. Madruga		11/13/2023 15:00
23K0684-02	DW - House	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 3.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #1

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Sample Results

Sample: DW - Barn
23K0684-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.26**	mmhos/cm	0.01	1		11/17/23 11:41	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1260**	umhos/cm	10.0	1		11/17/23 11:41	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0431
Nitrate Nitrogen as NO3N	**23.4**	mg/L	0.1	1	10	11/15/23 02:43	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:41	SM 4500-H+	H	BEK0478
pH	**7.8**	units	1.0	1		11/17/23 11:41	SM 4500-H+	H	BEK0478

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #1

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Sample Results
(Continued)

Sample: DW - House
23K0684-02 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.83**	mmhos/cm	0.01	1		11/17/23 11:42	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1830**	umhos/cm	10.0	1		11/17/23 11:42	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0431
Nitrate Nitrogen as NO3N	**24.6**	mg/L	0.1	1	10	11/15/23 03:03	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:42	SM 4500-H+	H	BEK0478
pH	**7.4**	units	1.0	1		11/17/23 11:42	SM 4500-H+	H	BEK0478

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #1

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0466										
Blank (BEK0466-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0466-BS1)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0466-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEK0466-DUP1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	20.3		0.1	mg/L		20.3			0.133	10
Duplicate (BEK0466-DUP2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	31.0		0.1	mg/L		31.3			1.08	10
Matrix Spike (BEK0466-MS1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	25.8		0.1	mg/L	5.000	20.3	110	90-110		
Matrix Spike (BEK0466-MS2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.1		0.1	mg/L	5.000	31.3	96.8	90-110		
Reference (BEK0466-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #1

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478										
Blank (BEK0478-BLK1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Blank (BEK0478-BLK2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.1		1.0	units						
Blank (BEK0478-BLK3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Duplicate (BEK0478-DUP1)		**Source: 23K0682-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	2.44		0.01	mmhos/cm		2.44			0.0205	10
pH	7.8		1.0	units		7.8			0.128	10
Electrical Conductivity umhos	2440		10.0	umhos/cm		2440			0.0205	10
Duplicate (BEK0478-DUP2)		**Source: 23K0690-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1.19		0.01	mmhos/cm		1.18			0.304	10
Electrical Conductivity umhos	1190		10.0	umhos/cm		1180			0.304	10
pH	7.8		1.0	units		7.8			0.128	10
Reference (BEK0478-SRM1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	562			umhos/cm	538.0		105	90-110		
Reference (BEK0478-SRM2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0478-SRM3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM4)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
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Received: 11/14/2023 15:45
Reported: 11/21/2023 09:40

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478 (Continued)										
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0478-SRM6)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM7)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM8)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0684

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples: 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT John Brasil Dairy #1
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW - Barn	11/13/2023	1500	3.5	ND
2	DW - House	11/13/2023	1600	2.9	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth			11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0684

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										